Colin Curtis - AI Systems and Software Engineering

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Professional Summary

Software engineering in the AI era is a complex and fast-moving profession where team-work and clear communication is as important as technical skill. Delivering great customer experience and value is the goal. In order to consistently reach that goal, cleanly-architectured systems and a focus on cost-effective practices are the vehicle.

Currently the technical lead for the AI Guardrails initiative at Fidelity Investments, building on three years as a core contributor to the Enterprise Generative AI platform serving 10,000+ users, hundreds of data scientists, and thousands of engineers.

Education

2012 - 2014 M.S. (Physics) at **North Carolina State University**
2007 - 2012 B.S. (Physics w/ Math Minor) at **Appalachian State University**

Skills and Interests

- Professional
  - C4 software system design
  - SOLID, YAGNI, DRY coding principles
  - AI/agentic pair programming
  - Modeling, Linear Algebra, Calculus, Statistics and Error Analysis
- Personal
  - Gardening and Cooking
  - Sci-fi and Fantasy
  - Mountain Biking, Skiing, Running and Soccer

Work Experience

Software Engineering and MLOps - Fidelity Investments April 2023 - Present

- Enterprise Data and AI (EDAI): AI Governance and Observability
- Tech Stack Highlights:
  - AWS: Bedrock, SageMaker, CloudWatch, ECR, S3, RDS, EKS, IAM, Cost Explorer, etc.
  - Azure: OpenAI (OAI) portal + OpenTofu (Terraform) model deployment/management
  - Python 3: botocore, flask, fastapi, pydantic, openai, streamlit, pandas, numpy, sqlalchemy, etc.
  - DevOps: Docker, Kubernetes, Splunk, Argo, Jenkins, Snowflake, Datadog, Jupyter NB, Open-Telemetry, Bruno

- Projects and Responsibilities:
  - Technical Lead - Guardrails, Enterprise AI Governance: LLM guardrails initiative safeguarding GenAI Gateway traffic — gather stakeholder requirements, translate them into architecture/design docs and JIRA tickets to parcel work across an engineering team, then manage technical integration timeline with partner teams
  - Lead Project , Internal Agentic AI Coding Assistant: Originated the concept and drove an initial proof of concept, then served as primary UI/UX stakeholder for 12+ months, shaping requirements and the developer experience into a working tool now being evaluated as an in-house alternative to paid third-party solutions
  - Lead reviewer across the AI/ML Solutions Engineering squad's primary repositories for 3 years (the 2 repos representing 90% of team work), ranking #1 or #2 in review volume, code suggestions, and discussion depth across 10+ GitHub activity metrics - patterns established through review have become the squad's de facto engineering standards
  - Mentored engineers from junior through principal level across the squad and was increasingly pulled into adjacent squads for expertise in MLOps, IaC, and LLM infrastructure
  - Onboarded GenAI Gateway to Fidelity's enterprise-wide API platform as one of only 8 core contributors, enabling secure LLM access for 10,000+ users; authored documentation supporting both internal teams and external consumers
  - Drove ML training/inference cost-savings initiative delivering five-figure monthly savings by implementing processes to detect and eliminate stagnant SageMaker deployments
  - Designed and shipped end-to-end automated testing pipeline for ML model deployment on AWS SageMaker, orchestrating parallel build/deploy/test/destroy workflows across Flask, DJL, and Triton model servers via Jenkins; supports hundreds of actively deployed models and up to 100 deployments per day. Resolved deep GPU/CUDA compatibility issues spanning NVIDIA drivers and SageMaker AMI configuration
  - Modularized MLOps model server build pipeline (Triton, DJL, custom) using Docker multi-stage builds, cutting repeat-build times by 10-60% per job and accelerating developer velocity for ˜150 engineers consuming the pipeline
  - Shipped enterprise JWT authentication enabling per-user identity, request attribution, and fine-grained access control for data science workflows, while maintaining backward compatibility with existing OAuth2 flows; serves 1,000+ client integrations and growing steadily
  - Owned Infrastructure-as-Code (OpenTofu/Terraform) for AWS Bedrock and Azure OpenAI deployments, managing dozens of customer accounts containing dozens of model deployments each, with the system onboarding ˜2-3 new accounts per week and accelerating
  - Eliminated heavyweight CSP SDK dependencies (boto3, etc.) from core codebases by implementing pure HTTP request signing and composition logic, reducing dependency footprint and resolving cascading version conflicts that had bogged down build and runtime performance
  - Rebuilt Jenkins automation for GenAI Gateway integration and performance tests; added weekly scheduled runs alongside manual execution and extended suites to auto-upgrade lower-environment deployments to latest versions before testing — catching regressions earlier with no manual intervention
  - Founding collaborator on an agentic AI coding CLI — analogous to Claude Code — that performs repo-scouting, generates plans, and applies in-IDE edits by routing through the GenAI Gateway; served as primary technical advisor to the data-scientist stakeholder on Gateway capabilities and constraints, and drove sustained feature ideation, usability feedback, and testing contributions across the tool's multi-year development

Senior Software Engineer - Garner Health August 2022 - February 2023

- Systems and Tech:
  - Python 3: connexion/flask, psycopg2, sqlalchemy, boto3, pytest, alembic, asyncio, behave
  - AWS: S3, VPC, IAM, RDS, ECR, Step Functions, Lambda, Transfer Family, Secrets Manager
  - Docker, Kubernetes, Terraform
- Projects and Responsibilities:
  - Designed and implemented a PGP decryption module for the client file ingestion pipeline, enabling secure processing of encrypted data deliveries and unblocking onboarding for new enterprise clients
  - Delivered Client Data Configuration Versioning system, providing rollback capability and audit trails for client-specific pipeline configurations across multi-tenant healthcare data infrastructure
  - Built comprehensive BDD test suite for RESTful API endpoints using the behave library, establishing living documentation that aligned engineering and product teams on expected system behavior
  - Authored Product Requirement Documents translating technical constraints into actionable product decisions for engineering and management stakeholders

Software Engineer - Actalent Services April 2019 - July 2022

- Systems and Tech:
  - Microservices architecture for Bridgestone's tire-design CAD web application
  - Python 3, React.js, numpy, logging, pytest, MongoDB
- Projects and Responsibilities:
  - Contributed to Bridgestone tire design CAD/analysis system written in Python 3 and React.js
  - Overhauled the CAD system UI to consolidate tire design engineer workflows into a single tool, introducing side-by-side design overlay comparison that eliminated the need to context-switch between separate applications
  - Refactored the automatic Excel report generation - tire engineers click a button in the CAD system and the Python back-end generates a downloadable report
  - Converted and upgraded FMAT, a tire image analysis tool, from MATLAB to Python 3 and introduced 'alpha-shapes' computational geometry technique to improve the boundary estimates of complex 2-D shapes

Junior Software Engineer - Renaissance Computing Institute June 2018 - March 2019

- NIH National Center for Advancing Translational Sciences (NCATS) - Biomedical Data Translator
- Wrote and deployed APIs using Marathon, Mesos, GitHub, Jenkins, Docker, nginx, and Python 3

Research Assistant, NCSU Physics May 2013 - April 2018

- Designed and coded ALAI, a MATLAB application for automating fractal analysis of nanoscopic images
  - Reduced the user's active analysis time, per image, by a factor of ˜50
- ANSYS Maxwell mesh-calculation to simulate 3-D electromagnetic fields
- LabVIEW software systems for instrument control and data collection
- Author/co-author of 3 peer-reviewed papers in experimental surface physics

Applied AI Engineering — Independent Work

Independent Study 2026

– Agentic systems, RAG, and MCP architecture: tool-use loops, multi-agent orchestration, human-in-the-loop patterns, agent isolation via tool-router enforcement (IAM analogue), retrieval failure modes, and context-bloat mitigation via the router pattern

– Eval frameworks (heuristic gates, LLM-as-judge, RAGAS-style metrics) and cost optimization (prompt caching, model tiering, batch processing) with real token-cost tradeoff math

Applied Project: B2B SaaS Churn Prevention System 2026

– Designed an end-to-end churn prediction system: time-boxed labeling, feature engineering with explicit handling of NULL semantics, class imbalance, and data leakage, driving a three-tier risk classifier and a RAG-backed agentic outreach system grounded in account-specific facts

– Built a multi-layer eval framework (heuristic gates, LLM-as-judge, golden test sets) and diagnosed two production incidents from monitoring data alone; proposed a treatment/control experimental design as the rigorous way to validate retention impact